

Your group has one paper, and it stays with you all day

Each group of three has been given one published economic evaluation from India. After each session, spend a few minutes applying what was just taught to **your** paper, using the prompts on its sheet and the **appraisal checklist**. By the report-back the checklist should be full and you should be able to say, in one sentence, whether you believe the paper's conclusion.

The eight papers were chosen to differ. Some are strong, some are not, and at least one calls itself something it is not. Do not assume that publication means quality, or that a paper from a well-known group is right.

THE EIGHT PAPERS

Group	Cluster	Paper	Access
1	Oncology and surgery	Thiagarajan et al. 2026, sentinel node biopsy in early oral cancer	open
2	Oncology and surgery	Guleria et al. 2025, laparoscopic IPOM for ventral hernia	open
3	Cardiology and medicine	Uy et al. 2021, surgery for rheumatic mitral valve disease	abstract only
4	Cardiology and medicine	Kaur et al. 2022, population screening for diabetes and hypertension	open
5	Critical care, anaesthesia, neonatology	Sharma et al. 2016, Kangaroo Ward Care for very low birth weight infants	open
6	Critical care, anaesthesia, neonatology	Patel et al. 2015, ward-based NIV for COPD respiratory failure	open
7	Paediatrics, infectious disease, nephrology	Gupta et al. 2022, peritoneal dialysis first	open
8	Paediatrics, infectious disease, nephrology	Srinivasan et al. 2024, childhood AML at a tertiary centre	open

REPORT-BACK, THE LAST SESSION

Groups report in **pairs**, chosen so that the two papers contrast. Each pair has six minutes: **two minutes per group**, then two minutes on what the contrast shows.

In your two minutes, give: (1) what the paper compared and what it concluded, in one sentence; (2) the one principle it handles **best**; (3) the one it handles **worst**, and whether that changes the conclusion; (4) your verdict: *would you act on this paper?*

Pair	Groups	Papers
1	1 and 2	Thiagarajan et al. 2026 and Guleria et al. 2025
2	4 and 3	Kaur et al. 2022 and Uy et al. 2021
3	5 and 6	Sharma et al. 2016 and Patel et al. 2015
4	7 and 8	Gupta et al. 2022 and Srinivasan et al. 2024

Each pair was chosen because the two papers differ in an instructive way. Working out how is part of the exercise.

Sentinel lymph-node biopsy guided neck dissection versus elective neck dissection in the management of early-stage oral cancer: a cost-utility analysis

Thiagarajan S, Sharda S, Chugh Y, Gupta N, Pramesh CS, Prinja S. *Cancer Med* 2026;15(2):e71571.

PMID 41644818 · PMC12875840 · doi:10.1002/cam4.71571

[Open access](#) Full text free at doi.org/10.1002/cam4.71571. Read it; this sheet only gets you started.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **Markov model** follows a hypothetical cohort of patients in India with early oral squamous cell carcinoma, tracking disease-free survival, recurrence and overall survival. Three strategies are compared: **sentinel lymph-node biopsy (SLNB) guided neck dissection**; **elective neck dissection (END) alone**; and **END with frozen section**.

Payer perspective, lifetime horizon, costs and QALYs discounted at 3%. One-way and probabilistic sensitivity analyses.

SLNB costs **INR 5,564** more per patient than END alone and **INR 2,507** more than END with frozen section, and gains **0.31** and **0.33** QALYs respectively. The reported incremental cost-utility ratios are **INR 8,088** and **INR 16,709** per QALY. Judged against one times GDP per capita, SLNB has a **94%** probability of being cost-effective. The authors support its inclusion in the government insurance scheme.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer? The authors name a specific scheme: which, and what would it need besides this paper?
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for?
Session 4 Costing your own service	SLNB needs a gamma probe, a radiotracer and nuclear medicine time. Look for these in the costs. Were capital costs annuitised?
Session 5 Outcomes, discounting and the ICER	Divide each incremental cost by its incremental QALYs. Do you get the ratios the paper reports? If not, find out why.
Session 6 Thresholds, opportunity cost, budget impact	They judge against one times GDP per capita . Would the verdict change against India's own estimate from Session 3?
Session 7 Modelling in half an hour	Name the health states. How long is a cycle? Where did the recurrence probabilities come from?
Session 8 Handling uncertainty	Which kinds of uncertainty were tested, and how? What is the probability that the conclusion is right?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?

Review for cost-effectiveness analysis of laparoscopic Intra-peritoneal Onlay Mesh for ventral hernia repair in Indian settings

Guleria C, Kumar D, Sahoo KC. *Cost Eff Resour Alloc* 2025;23(1):27.

PMID 40495189 · doi:10.1186/s12962-025-00638-4

[Open access](#) Full text free at doi.org/10.1186/s12962-025-00638-4. Read it; this sheet only gets you started.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **systematic review and meta-analysis of 10 randomised trials** (1,204 patients) comparing **laparoscopic intra-peritoneal onlay mesh (IPOM)** repair with **open ventral hernia repair**, combined with a cost-effectiveness analysis from the **Indian health system perspective** over five years. Costs come from the National Health System Cost Database.

Hernia recurrence, the primary outcome, did not differ (risk ratio **1.28**, 95% CI 0.81 to 2.04). Wound infection was lower with the laparoscopic repair (risk ratio **0.31**, 0.18 to 0.54).

The ICER is **INR 5,023 per wound infection averted**, set against a threshold of **INR 2,14,000** based on India's per-capita income. In a probabilistic analysis of 10,000 simulations the laparoscopic repair was cost-effective in **54.0%**. The authors conclude it is not clinically effective for recurrence and less likely to be cost-effective.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer? Look at the authors' affiliations.
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for? Is five years long enough for hernia recurrence?
Session 4 Costing your own service	Where did the costs come from? Bottom up or top down? Whose costs are counted, and whose are not?
Session 5 Outcomes, discounting and the ICER	The primary outcome was recurrence. Which outcome is in the ICER? Why might that choice matter?
Session 6 Thresholds, opportunity cost, budget impact	The threshold is per-capita income. What unit of health is that threshold meant to buy? Is it the unit in this paper's ICER?
Session 7 Modelling in half an hour	Is there a model at all, or is this a calculation on pooled trial results? Does it need one?
Session 8 Handling uncertainty	54% of simulations favoured the laparoscopic repair. The authors call it 'less likely to be cost-effective'. Do you agree?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?

Cost-utility analysis of heart surgeries for young adults with severe rheumatic mitral valve disease in India

Uy J, Ketkar AG, Portnoy A, Kim JJ. *Int J Cardiol* 2021;338:50-57.

PMID 34090957 · doi:10.1016/j.ijcard.2021.05.059

Not open access Work from this sheet and the abstract. **Not reported in the abstract** is an acceptable answer to a prompt, and worth saying out loud.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **Markov model** of a hypothetical cohort of **20-year-olds** in India with severe rheumatic mitral valve disease, from an **Indian public payer perspective**, estimating lifetime costs and QALYs.

Three strategies: **valve repair**; **mechanical valve replacement**; **bioprosthetic valve replacement**. The comparator is a **standard-of-care mix** approximating Indian practice: 32% repair, 33% mechanical, 35% bioprosthetic.

Results per patient (US dollars): repair **\$2,530, 9.7 QALYs**; standard of care **\$2,990, 8.7 QALYs**; mechanical **\$3,220, 6.2 QALYs**; bioprosthetic **\$3,190, 10.1 QALYs**.

Repair is cheaper and more effective than both standard of care and mechanical replacement. Bioprosthetic replacement against repair costs **\$1,590 per QALY**, which the authors say may be cost-effective against India's GDP per capita (\$2,005). They conclude that repair is optimal, mechanical replacement should not be recommended for this group, and bioprosthetic replacement may be cost-effective where repair quality is not assured, newer valves are used, or valve prices fall.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer? Where are the authors based?
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for? Is a 'standard-of-care mix' a real comparator?
Session 4 Costing your own service	Costs are in US dollars. What would you want to know about the conversion and the cost year?
Session 5 Outcomes, discounting and the ICER	Put the four strategies in order of QALYs. Which are dominated? Recompute the bioprosthetic vs repair ratio.
Session 6 Thresholds, opportunity cost, budget impact	Against what threshold is repair 'optimal' rather than bioprosthetic replacement? Try India's own estimate from Session 3.
Session 7 Modelling in half an hour	A Markov model over a lifetime from age 20. What has to be assumed about valve durability and re-operation?
Session 8 Handling uncertainty	The abstract does not describe the sensitivity analyses. What would you most want tested?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?

Cost-effectiveness of population-based screening for diabetes and hypertension in India: an economic modelling study

Kaur G, Chauhan AS, Prinja S, Teerawattananon Y, Muniyandi M, Rastogi A, Jyani G, Nagarajan K, Lakshmi P, Gupta A, Selvam JM, Bhansali A, Jain S. *Lancet Public Health* 2022;7(1):e65-e73.

PMID 34774219 · doi:10.1016/S2468-2667(21)00199-7

Open access Full text free at doi.org/10.1016/S2468-2667(21)00199-7. Read it; this sheet only gets you started.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **hybrid decision tree and Markov model** of a cohort aged 30, over a **lifetime**, from a **societal perspective** that includes health-system and out-of-pocket costs, discounted at 3%. Population-based screening at different ages, frequencies and tests is compared with **current practice**, where diabetes and hypertension are detected opportunistically.

Costs come from primary data in Haryana and Tamil Nadu, the National Health System Cost Database, the Costing of Health Services in India study, national survey data and insurance package rates. Quality of life was measured with **EQ-5D-5L in 962 patients**. Funded by the Department of Health Research.

Judged against one times GDP per capita, **no screening strategy was cost-effective at current levels of health-care use**; even screening once in 20 years had only a 60% probability. But if **20%** of newly diagnosed patients were treated through Health and Wellness Centres, screening people aged 30 to 65 every three or five years became cost-effective, and at **70%** annual screening became cost saving.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer?
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for?
Session 4 Costing your own service	This paper collected its own cost and quality-of-life data. What did that buy that a database could not?
Session 5 Outcomes, discounting and the ICER	Find the ICER for one strategy you think is realistic. Is the utility weight from the Indian value set?
Session 6 Thresholds, opportunity cost, budget impact	Which threshold? And the programme is population-wide: what would its budget impact be, and does the paper say?
Session 7 Modelling in half an hour	Why a decision tree <i>and</i> a Markov model? What does each part do?
Session 8 Handling uncertainty	The verdict flips on one number. Which is it, and is it clinical?

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Group 5

To compare cost effectiveness of 'Kangaroo Ward Care' with 'Intermediate intensive care' in stable very low birth weight infants (birth weight < 1100 grams): a randomized control trialSharma D, Murki S, Oleti TP. *Ital J Pediatr* 2016;42:64. PMID 27412638 · doi:10.1186/s13052-016-0274-3

Open access

Full text free at doi.org/10.1186/s13052-016-0274-3. Read it; this sheet only gets you started.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **randomised trial** in a tertiary neonatal unit in Hyderabad. **141** stable very low birth weight infants (71 and 70) were randomised to **Kangaroo Ward Care**, moving to the kangaroo ward with the mother at 1,150 g, or **Intermediate Intensive Care** in an incubator or warmer until 1,250 g.

Hospital costs were estimated top down and parents' out-of-pocket costs bottom up, from randomisation to discharge.

Mean cost was **INR 41,592** per infant with Kangaroo Ward Care and **INR 75,389** with intermediate care, a saving of about **INR 33,800 (US\$512)** per infant. NICU and hospital stays were shorter. The authors conclude that early transfer to the kangaroo ward is a cost-effective intervention.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer?
Session 2 What economic evaluation is	Drummond's two questions: are both costs <i>and</i> consequences compared? Which box is it in?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for? When does the horizon end?
Session 4 Costing your own service	Top down for the hospital, bottom up for the parents. What is the difference, and what does each risk missing?
Session 5 Outcomes, discounting and the ICER	What is the outcome measure? Is there an ICER? If the babies do equally well, what kind of analysis is this, and what must be shown first?
Session 6 Thresholds, opportunity cost, budget impact	Is any threshold used? Could one be?
Session 7 Modelling in half an hour	Is there a model? Would a longer horizon need one?
Session 8 Handling uncertainty	Which kinds of uncertainty were tested, and how? What is the probability that the conclusion is right?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?

Group 6

Cost-effectiveness of noninvasive ventilation for chronic obstructive pulmonary disease-related respiratory failure in Indian hospitals without ICU facilitiesPatel SP, Pena ME, Babcock CI. *Lung India* 2015;32(6):549-556.

PMID 26664158 · PMC4663855 · doi:10.4103/0970-2113.168137

[Open access](#) Full text free at doi.org/10.4103/0970-2113.168137. Read it; this sheet only gets you started.**WHAT THE PAPER SAYS IT DID, AND FOUND**

A **decision-analytic model** comparing **ward-based non-invasive ventilation (NIV)** with **standard treatment** against **standard treatment alone** for COPD-related respiratory failure in Indian hospitals that have no ICU. All parameters come from the published literature. **Societal perspective, lifetime horizon**, future costs discounted at 3%, costs in 2012 US dollars. One-way, two-way and probabilistic sensitivity analyses.

NIV gave 17.7% more survival, cost **\$101** more and gained **1.67 QALYs**. The paper reports cost per QALY of **\$78** for standard care ($\$535.02 \div 6.82$) and **\$75** for NIV ($\$636.33 \div 8.49$), and an ICER of **\$61 per QALY**, far below India's GDP per capita (\$1,489). NIV was preferred in 100% of simulations at a willingness to pay above \$250.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer? Where are the authors based?
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for?
Session 4 Costing your own service	Every parameter is from the literature. Which of them would you most want replaced with Indian data from your own ward?
Session 5 Outcomes, discounting and the ICER	The paper reports \$78 and \$75 per QALY, and also \$61. Which of these is the ICER, and what are the other two? Are QALYs discounted?
Session 6 Thresholds, opportunity cost, budget impact	Against GDP per capita the answer is obvious. Would any plausible threshold change it? Then what is the real question for a hospital?
Session 7 Modelling in half an hour	What kind of model? What happens to a patient after discharge in it?
Session 8 Handling uncertainty	100% probability of being cost-effective. What does PSA <i>not</i> capture here?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?

Peritoneal dialysis-first initiative in India: a cost-effectiveness analysis

Gupta D, Jyani G, Ramachandran R, Bahuguna P, Ameel M, Dahiya BB, Kohli HS, Prinja S, Jha V. *Clin Kidney J* 2022;15(1):128-135. PMID 35035943 · PMC8757426 · doi:10.1093/ckj/sfab126

Open access Full text free at doi.org/10.1093/ckj/sfab126. Read it; this sheet only gets you started.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **Markov model** with one-year cycles compares starting kidney replacement therapy with **peritoneal dialysis (PD first)** against starting with **haemodialysis (HD first)** in people with kidney failure in India, over their lifetime, with complications such as peritonitis, vascular access problems and blood-borne infections.

Two scenarios: a **real-world** one, using current costs and patterns of use; and a **public programme** one, following the Pradhan Mantri National Dialysis Programme. Health system costs, patients' out-of-pocket spending and indirect costs are all included; 3% discounting.

Mean QALYs were **3.3 with PD and 1.6 with HD**. From a societal perspective PD first is **cost-saving** in both scenarios. If only costs directly attributable to care are counted, PD first is cost-effective only if PD consumables cost **INR 70 per unit**, 65% below the recommended price of INR 200. The authors conclude the government should negotiate the price of PD consumables under the programme.

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer? The conclusion is addressed to someone specific: who?
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for? Is 'HD first' what happens now?
Session 4 Costing your own service	Health system, out-of-pocket and indirect costs are all in the base case. Which of those does the Indian Reference Case put in the base case, and which only in sensitivity analysis?
Session 5 Outcomes, discounting and the ICER	Where do the quality-of-life weights come from? Is it the Indian value set?
Session 6 Thresholds, opportunity cost, budget impact	Which threshold? And the conclusion is a price, not a verdict: what kind of answer is that?
Session 7 Modelling in half an hour	Name the health states and the cycle length. Which complications are modelled?
Session 8 Handling uncertainty	The answer changes with one perspective choice and one price. Which, and by how much?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?

Cost-effectiveness of treating childhood acute myeloid leukemia at a tertiary care center in North India

Srinivasan S, Bolous NS, Batra A, Bharti S, Singh N, Shaikh T, Yadav A, Kanwar V. *Pediatr Blood Cancer* 2024;71(11):e31242. PMID 39126354 · doi:10.1002/pbc.31242

[Open access](#) Full text free at doi.org/10.1002/pbc.31242. Read it; this sheet only gets you started.

WHAT THE PAPER SAYS IT DID, AND FOUND

A **retrospective study** of children under 15 with acute myeloid leukaemia treated at one tertiary cancer centre in North India. Outcomes, complications, deaths and costs were taken from the electronic medical record and the hospital database.

Of **59** children, **43** received intensive therapy. Treatment-related mortality reached **30%**, mostly from sepsis; abandonment and limited access to bone marrow transplantation were also problems. Three-year event-free survival was **24.5%** and overall survival **27.9%** in those treated intensively.

The total cost per newly diagnosed child treated with curative intent was **US\$4,454**. Cost-effectiveness was measured as cost per **DALY averted**, which came to **24% of GDP per capita**; the authors judge treatment cost-effective even against a stringent threshold, and call for reducing treatment-related deaths.

A **published Comment** on this paper appeared in the same journal (Sra, Bakhshi, Ganguly, *Pediatr Blood Cancer* 2025;72(1):e31305, PMID 39228042).

YOUR PROMPTS THROUGH THE DAY

After	Ask of your paper
Session 1 Where economic evidence enters policy	Who funded or commissioned it, and who in India could act on its answer?
Session 2 What economic evaluation is	Which of Drummond's six boxes is it in? If it is a full evaluation, which of the four types? What is the comparator, even if the paper does not name one?
Session 3 Framing the question	Is the comparator what actually happens now? Are the perspective and time horizon the ones India asks for? How are children who abandoned treatment counted?
Session 4 Costing your own service	Costs come from the hospital database. Whose costs are these, and what does a family pay that is not in them?
Session 5 Outcomes, discounting and the ICER	DALYs, not QALYs. Can this result be set beside any other paper in the pack? What does the Reference Case allow for children?
Session 6 Thresholds, opportunity cost, budget impact	Cost per DALY averted at 24% of GDP per capita. Which threshold is 'stringent', and would it pass India's own estimate?
Session 7 Modelling in half an hour	Is there a model, or is this observed data? What happens to survival after three years?
Session 8 Handling uncertainty	Is any uncertainty reported? Then read the published Comment: did its authors find what you found?

The summary paraphrases the published abstract and is not a substitute for it. Citation checked against the MEDLINE record, September 2026. By the report-back: would you act on this paper, and why?